Requirements Elicitation (User Stories)

Selection of Requirement Stories

Transformation of Stories into CQs

Requirements were elicited in the form of user stories derived from real-world survey scenarios. These stories describe how domain experts design and use surveys, and they serve as the basis for identifying modelling requirements.

Instead of selecting individual stories in a strictly iterative manner, requirement stories were organised into **thematic clusters** (e.g., Survey Structure, Survey Components, Questions and Answers). Each cluster defines a coherent modelling scope and corresponds to a conceptual module of the ontology.

Requirement story card: Survey Structure and Organization

User story:

A social science researcher, Giorgio Rossi, is conducting a study aimed at analysing the correlation between second-generation immigrants and social disadvantage. The study investigates several socio-economic and demographic variables, including language proficiency, educational background, employment status, political orientation, and economic conditions.

To support a robust longitudinal analysis, Giorgio plans to administer the survey periodically over a five-year period, collecting data from different population groups characterised by diverse socio-demographic profiles. The repeated administration of the survey allows the researcher to analyse changes over time and compare trends across different groups of participants.

During the course of the study, the survey evolves iteratively: new versions are introduced to improve the quality of data collection, refine specific questions, or adapt the questionnaire to emerging research requirements. As a consequence, it becomes necessary to manage relationships between survey versions, track their derivation history, and represent the time periods during which each version is used.

Furthermore, Giorgio needs to associate each survey with the corresponding research study, define recurring collection periods, and specify the start and end dates of each survey administration cycle in order to support the management of longitudinal research activities.

Story decomposition

The researcher has, i.e.

- a research study focused on analysing correlations between social disadvantage and multiple socio-economic and demographic variables.
- a set of surveys administered periodically within a longitudinal research setting.
- multiple evolving versions of the same survey, reflecting iterative refinements and methodological adaptations.
- temporal knowledge related to survey administration, including recurring collection periods and specific start and end dates.
- explicit associations between surveys and the research studies to which they belong.
- versioning relationships describing how survey versions derive from previous ones.

Extracted CQs

- What surveys are associated with a given study? → **CQ1**
- What are the different versions of a given survey? → **CQ2**
- From which previous version was a given survey version derived? → **CQ3**
- What is the recurrence pattern of a longitudinal survey? → **CQ4**
- What are the start and end dates of each data collection period? → **CQ5**

User Story

Giorgio Rossi begins designing his survey by reviewing existing scientific studies and survey instruments in order to identify the most relevant research variables and suitable answer options. Based on this analysis, he structures the questionnaire into different survey components, such as groups, questions, and answer options, defining their logical organisation and ordering within the survey.

Each component is uniquely identified and enriched with descriptive attributes represented as name–value pairs, enabling a flexible and configurable representation of the questionnaire structure. Giorgio also establishes explicit relationships between questions and the variables they are intended to measure, ensuring consistency between the survey design and the research objectives.

The questionnaire evolves iteratively throughout the design process: components, variables, and configurations are progressively refined and reorganised in response to methodological requirements and preliminary evaluations.

Story decomposition

The researcher has, i.e.

- knowledge of existing surveys and scientific studies used to identify relevant research variables.

- a structured survey composed of different components, such as groups, questions, and answer options.
- an organisation of survey components based on a defined logical and sequential order.
- uniquely identifiable survey components associated with identifiers.
- descriptive knowledge about survey components represented through attributes expressed as name–value pairs.
- explicit associations between questions and the variables they are intended to measure.
- a set of variables collected through the survey.
- iterative refinements of survey components and configurations during the questionnaire design process.

Extracted CQs

- What components does a survey contain? → CQ6
- What is the order of components within a survey? → CQ7
- What are the identifiers associated with each survey component? → CQ8
- What attributes does a specific component have? → CQ9
- What is the name and value of a component attribute? → CQ10
- What variable name is associated with each question? → CQ20
- What variables are collected in a survey? → CQ21

User Story

Giorgio Rossi defines the questions of his survey, identifying how they should be grouped, presented, and ordered in order to maximise the quality and reliability of the collected data. He determines the conditions under which specific questions or groups of questions should be displayed, defines different question types, and specifies the corresponding answer options and default values where necessary.

Some questions are further articulated through subquestions to model more complex structures. Each question is explicitly associated with one or more research variables to ensure consistency between the questionnaire and the study objectives.

After constructing an initial version of the questionnaire, Giorgio tests it with a small group of participants in order to evaluate clarity, usability, and coherence. Based on the collected feedback, he refines some questions and question groups, adjusts answer options and ordering, and finally defines the final version of the questionnaire.

Story decomposition

The researcher has, i.e.

- a set of survey questions designed to capture specific research variables.
- an organisation of questions into coherent question groups.
- knowledge about the conditions under which questions or groups should be displayed.
- different question types determining how responses are collected.
- a set of answer options associated with questions or question groups.
- default values assigned to specific questions where necessary.
- complex question structures including subquestions.
- an ordering of answer options defining how they are presented to respondents.
- explicit associations between questions and research variables.
- iterative refinements derived from user testing and feedback collection.

Extracted CQs

- What questions belong to a specific question group? → **CQ11**
- What is the condition for displaying a particular question or question group? → **CQ12**
- What answer options are available for a given question? → **CQ13**
- What answer options are available for a given question group? → **CQ14**
- What is the default value for a specific question? → **CQ15**
- What type does a question have? → **CQ16**
- What variable is associated with a question? → **CQ17**
- Does a question have subquestions, and if so, what are they? → **CQ18**
- What is the order of answer options for a question or question group? → **CQ19**

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